

DEPOSITS AND PAYMENT SERVICES

(ROLE OF DEPOSITS IN BANKS' BS AND P&L, TYPES OF DEPOSITS, HISTORY OF PAYMENT SERVICES, SEPA, PSD2, FINTECH INNOVATIONS, DEPOSIT INSURANCE)

Source:

- *Introduction to banking book*, Barbara Casu

A **bank** is a financial intermediary whose core activity is to *provide loans to borrowers and to collect deposits from savers* (collect funds (deposits) from units in surplus and lend funds (loans) to units in deficit).

Modern banks offer a wide range of financial services, including:

- Payment services
- Deposit and lending services (*bulk of their money comes from deposits*)
- Investment, pensions and insurance services
- E-banking

Characteristics of deposits: *small size, low risk and high liquidity*.

Transformation function: banks match the different needs of borrowers and lenders by *transforming deposits into loans* which are of larger size, higher risk and illiquid. Bank liabilities (i.e., deposits) are accepted as a means of exchange. Banks are the only intermediaries that can vary the level of deposits and can create and destroy credit.

The monetary function of bank deposits is often seen as one of the main reasons why deposit-taking institutions (DTIs) are subjected to *heavier regulation and supervision* than their non-deposit-taking institution (NDTI) counterparts (such as insurance companies, pension funds, investment companies, finance houses and so on).

Role of deposits

a) Balance Sheet (BS):

Deposits and payment services

Role of deposits, types, history of payment services, SEPA, PSD2, fintech innovations, deposit insurance

- The *balance sheet* provides information about the bank's financial position at the end of the accounting period.
- **External sources of financing** held on the *liabilities side of a bank's balance sheets*; they represent funds that the bank owes to its customers. These deposits are *retail deposits, company deposits* and *interbank deposits*.
- Have *small size, low risk* and *higher liquidity* than equity or bonds. They are financial claims from the side of depositors.
- Deposits are a *primary source of funding* for banks providing the necessary liquidity for banks to conduct lending and investment activities.
- Bank *liabilities* tend to have *shorter maturities than assets*. This mismatch comes from the *different requirements of borrowers and lenders*: depositors want to lend for short time at the highest possible return, while borrowers want cheap loans for long periods. Matching this need is done through the *transformation function* of a bank: transforming deposits into loans.
- Deposits are *contractual agreements that offer high liquidity and low risk*: these are financed by relatively illiquid and higher risk assets (e.g., loans) on the assets side of the balance sheet.
- The bank also has *deposits on the assets side* of its BS by placing deposits with the FED or other banks.

Table 9.1 Simplified commercial bank balance sheet

Assets	Liabilities
Cash	Deposits: retail
Liquid assets	Deposits: wholesale
Loans	
Other investments	
Fixed assets	
	Equity
	Other capital terms
Total assets	Total liabilities and equity

Once deposits have been collected, a bank can use them for:

Deposits and payment services

Role of deposits, types, history of payment services, SEPA, PSD2, fintech innovations, deposit insurance

- *lending*: make loans to individuals and businesses; interest earned on loan is greater than interest paid on deposits
- *investment*: invest the money from deposits into securities that usually earn higher interest rates; additional income to pay deposits
- *liquidity*: source of liquidity; a steady flow of funds to meet these demands (ex. withdrawal)
- *regulation*: certain level of reserves; against deposits to ensure they can meet customer demands

b) Profit and Loss Statement (P&L):

- The P&L statement measure a bank's performance between two year-end balance sheets.
- **Interest income** is the income generated on bank's assets like loans, securities (investments) or deposits lent out to other institutions. **Interest expense** is the interest paid on all interest-bearing liabilities like deposit accounts, CDs, short-term borrowing and long-term debt. The difference between interest income and interest expenses is the *net interest income* (NII).
- **Non-interest income** is generated by fees, commissions and trading income, and includes fees and service charges related to deposits (account maintenance fee, overdraft fee, ATM fee etc).
- **Non-interest expenses** contain costs related to deposits: the cost of maintaining and servicing deposit accounts, including administrative expenses and customer service costs.

Types of deposits

(Source: <https://corporatefinanceinstitute.com/resources/wealth-management/deposit/>
<https://www.investopedia.com/terms/d/deposit.asp#toc-types-of-deposits>)

The composition of deposits, including the proportion of low-cost deposits (e.g., checking and savings accounts) versus higher-cost deposits (e.g., time deposits), can impact a bank's profitability and liquidity position.

Demand Deposit vs. Term Deposit

- A demand deposit account and a term deposit account are both types of financial accounts, differing in terms of *accessibility or liquidity*, and in the *amount of interest* that can be earned on the deposited funds.
- Basically, a DDA allows funds to be accessed anytime, while a TD account restricts access to funds for a predetermined period.
- Typically, deposits that can be withdrawn on demand pay lower rates than those deposited in the bank for a set period.

1. Demand deposit (DDA):

- allows funds to be accessed *anytime*
- consumers can *deposit money and withdraw it as needed* (using various methods such as bank cards at ATMs, checks, or over-the-counter withdrawal slips)
- While some banks may charge *monthly fees* for maintaining a current account, they may waive these fees if certain conditions are met, such as setting up direct deposit or making a minimum number of monthly transfers to a savings account.

a) Current or checking accounts / Sight deposit

- Consumers deposit money, and the deposited money can be withdrawn or spent as the account holder desires
- withdraw funds using bank cards or checks
- typically pay no (or very low) rates of interest and are used mainly for payments
- banks may charge monthly fees for current accounts, but they may waive the fee if the account holder meets other requirements, such as setting up direct deposit or making a certain number of monthly transfers to a savings account.
- Goal is to hold your money in a *secure* place for the *short term*

b) Savings Account:

- Money is tied up for a longer period
- Typically *higher interest* rates and *less liquidity* than checking accounts, income is taxable
- Have some *limitations* on how often you can withdraw funds, but generally offer exceptional flexibility
 - Banks may charge fees for early withdrawal of funds from savings accounts.
- Some savings accounts may incur a monthly fee if a minimum balance is not maintained, or a certain number of deposits are not made.

c) Money Market Account:

- A mix of current account and saving account
- *Higher minimum level* required and *higher interest rate* than on saving account, income is taxable
- limited check-writing ability, limited number of transfers/transactions
- the *deposits are invested* by the bank in low-risk assets (CDs, US Treasuries, commercial papers)

2. Term deposit / Time deposit:

- Restricts access to funds for a predetermined period
- Usually pay a fixed interest rate
- Money in the account earns interest until a set maturity date
- These interest-earning accounts offer higher rates than savings accounts
- The money cannot be withdrawn before maturity without a penalty and withdrawals often require written notice in advance

a) Certificate of deposit (CD)

- Typically *fixed rate and higher interest rate* than savings accounts, and money tied up for a *specified period of time* (from 1 month to 5 years)
- Early withdrawal is penalized
- At maturity either withdraw the money or reinvest
- Can be sold in the market whenever an individual/firm needs cash: depositors, instead of waiting until maturity of the securities, may sell them in the market to regain the cash - enhances the liquidity of the depositors' funds (in the broadest sense)

Advantages of deposits:

Deposits and payment services

Role of deposits, types, history of payment services, SEPA, PSD2, fintech innovations, deposit insurance

- *Safety and low risk*: banks are usually insured by governments
- *Convenient*: easy to open and maintain, deposit or withdraw anytime, anywhere (online)
- *Guaranteed return*

Disadvantages of deposits:

- *Low return*
- *Inflation risk*
- *Liquidity risk*: account type dependent
- *Fees*

Deposit Insurance

Deposit insurance is a guarantee to depositors that all or part of their deposits will be paid in the event of a bank failure. The primary goal is to *prevent bank runs* and *protect consumers* from losing their savings.

United States:

- The Federal Deposit Insurance Corporation (FDIC) guarantees deposits for at least \$250,000 per depositor, per insured bank.

United Kingdom:

- The Financial Services Compensation Scheme (FSCS) administers deposit protection arrangements introduced under the Financial Services and Markets Act 2000 (FSMA) covering deposits up to £85,000 per depositor, per authorized firm.

Europe:

- Building on national schemes: EDIS complements existing national Deposit Guarantee Schemes (DGS) to ensure protection for deposits up to €100,000 across the EU.
- European Deposit Insurance Scheme (EDIS): proposed in November 2015 as part of completing the banking union and deepening the economic and monetary union. Has the scope to supplement and overcome the weaknesses of national DGSs.
- Uniform Coverage: Aims to provide consistent insurance cover for deposits in the euro area, reducing reliance on national schemes and bolstering depositor confidence.
- Application to All Banks: Covers deposits below €100,000 in all banks within the banking union, intervening in cases of bank insolvency or resolution.

Deposits and payment services

Role of deposits, types, history of payment services, SEPA, PSD2, fintech innovations, deposit insurance

- Travel and Entertainment Cards: Provide payment facilities with repayment deferred until the end of the month, requiring full repayment without rollover.
- Smart, Memory, or Chip Cards: Incorporate microprocessors or memory chips for added security and various functions, commonly used in Europe for card payments and store value transactions.

For small businesses, accessing basic payment services is crucial. Small firms use their business accounts to deposit cash and checks at bank branches and withdraw cash from ATMs. In the UK, most business transactions involve checks, direct debits, and standing orders, with cash and card payments being less common, making up only about 5% of all business payments.

History of Payment services

- **Barter System (Prehistoric times):** In ancient times, *people exchanged goods and services directly*, known as the barter system. People traded things they had for things they needed. While effective for simple transactions, it lacked standardization and posed challenges for complex trade.
- **Commodity money (3,000 BC):** using valuable objects like seashells, cattle, or precious metals (gold, silver) for transactions.
- **Coins (7th century BC):** The introduction of *coins* (precious metals) led to a standardized medium of exchange. These coins were portable, durable, and served as a universal unit of value.
- **Paper Money (7th century):** initially as promissory notes, after that banknotes.
- **Banking system (Medieval period):** initially consisted of banking families, then central banks emerged. This era introduced checks, bills of exchange and early credit systems.
- **Payment Cards (Mid-20th century):** emergence of credit cards, revolutionizing payment methods. Initially issued by individual stores and later by banks, credit cards allowed consumers to make purchases on credit, with payments settled at a later date.
- **Electronic Payments (1970s-1980s):** The digital age ushered in a new era of electronic payments. Electronic fund transfers (EFTs) enabled instant transactions, eliminating the need for physical currency.

- **Online banking and E-commerce payments (1990s):** Internet banking allowed users to view balances and transfer funds online, and services like PayPal emerged, revolutionizing online payments.
- **Mobile Payments and Digital Wallets (2000s):** the rise of smartphones facilitated mobile banking and payments: conveniently make payments, transfer funds, and manage their finances on smartphones. Contactless payment methods, such as NFC-enabled cards and mobile wallets (ApplePay, GooglePay or SamsungPay) offer speed and convenience.
- **Cryptocurrencies and Blockchain (2009 onwards):** The emergence of cryptocurrencies like Bitcoin introduced decentralized digital currencies built on blockchain technology. Offering privacy, security, and borderless transactions.
- **Embedded finance and BNPL (2020s):** payment services are integrated directly into non-financial platforms (e.g., Uber's in-app payments). When using "Buy Now, Pay Later" you can make a purchase at a participating retailer and opt for buy now, pay later at checkout (no interest, no additional charges).

Payment Services

Income sources of a financial institution include revenue generated from transactional services, such as payment services. Banks offer important payment services enabling people and businesses to transfer money securely. These services are vital for everyday transactions, including buying goods, paying bills, and transferring funds between accounts. Banks ensure these processes run smoothly. Both paper-based and electronic payment methods are available, contributing to a well-functioning financial system.

- **Cheques:** used to instruct a bank to pay a specific amount of money from one person's account to another.
- **Cashless payments**
 - *Credit Transfers:* Direct transfer of funds from the payer's bank account to the beneficiary's account, commonly used for invoices or advance payments.
 - *Standing Orders:* Fixed, regular payments instructed by the account holder to the bank for recurring transfers.

Deposits and payment services

Role of deposits, types, history of payment services, SEPA, PSD2, fintech innovations, deposit insurance

- *Direct Debits*: Initiated by the supplier (payee), with variable payment amounts and fixed or variable debiting times, often used for utility bill payments. You allow a specific organization (payee) to collect payments from your account.
- ***Card payments***: including credit cards, debit cards, and cheque guarantee cards, facilitating electronic or paper payments.
 - *Credit Cards*: Offer pre-arranged credit limits for purchases, with interest charges if not paid in full by the due date.
 - *Debit Cards*: Allow direct withdrawals from bank accounts and cash access via ATMs.

Regulation

- **Payment Service Directive I (PSD1)** – 2007: standardize payment services and enhance consumer protection, transparency, and competition in the EU's payment industry. It was phased out in 2018.
- **Development of Single European Payment Area – SEPA**: An initiative to create a unified payment area across European countries, facilitating seamless euro-denominated transactions.

SEPA (Single Euro Payments Area)

- European Union project that aims to create an *integrated market for all electronic euro payments within Europe*.
- Ensures *fast, safe and cost-effective (cashless) euro payments* comparable to national payments in the EU and EEA
- *Harmonizes payment systems by standardizing payments across participating countries* eliminating differences between national and cross-border payments; common standards for credit transfers, direct debits, and card payments, making it easier to initiate and receive euro payments across borders.
- *Speed of processing*: no longer than one business day for electronic payment orders; two business days for paper-based payment orders

PSD2 (Payment Services Directive 2)

Deposits and payment services

Role of deposits, types, history of payment services, SEPA, PSD2, fintech innovations, deposit insurance

- It updates and enhances PSD1, and came into force in 2018
- Aim is to *enhance security in payments across Europe*: protect consumers, foster competition and innovation, and reinforce security in the payments market.
- *Open Banking*: allowing third-party providers to access customer financial information with their consent, fostering innovation in financial services.
- Regulation of Payment Initiation Services (PIS) and Account Information Services (AIS).
- Enhancing security with *Strong Customer Authentication (SCA)* which makes electronic payments more secure; requires multi-factor verification (more steps beyond just a password), such as fingerprint recognition or a one-time code.
- Improved Consumer Protection through *Liability for Unauthorized Payments* (consumers' liability for unauthorized transactions is limited) and *Refund Rights for Direct Debits* (consumer's right to a refund for unauthorized or incorrect direct debits)

FinTech innovations

- Fintech refers to the *integration of technology into offerings* by financial services companies to improve their use and delivery to consumers.
- Companies in the finance industry that use fintech have expanded financial inclusion and use technology to cut down on operational costs.
- Fintech funding is on the rise, but regulatory problems exist.

Biometric Authentication:

- Biometric technology enhances identity verification and ensures secure financial transactions to prevent fraud: fingerprint, facial recognition, and iris scanning are increasingly used for secure and convenient payment authorization, adding another layer of security
- They can be used in multi-factor verification ensuring secure digital transactions.

Mobile Wallets and Contactless Payments:

- Using smartphones and wearable devices for secure and convenient contactless payments (Apple Pay, Google Pay or Samsung Pay)

Blockchain & Cryptocurrencies:

- *Blockchain* is a digital record-keeping system, where transactions are bundled together into blocks, and these blocks are linked chronologically using cryptography, making them very secure
- *Cryptocurrencies* are digital assets that live on blockchains. They can be used to pay for things or trade like other investments. They are secure because they rely on the blockchain for verification and record-keeping, instead of a central authority like a bank.

Artificial Intelligence:

- AI, machine learning, and robotic process automation improve risk management and customer experiences.
- Chatbots and machine learning enhance customer service and predict client behavior.

Embedded Finance:

- Integration of financial services into non-financial platforms enhances customer experience and revenue streams.
- Companies like Amazon offer financial services like Amazon Pay and Amazon Lending within their ecosystems.
- Embedded finance blurs the lines between traditional banking and everyday activities.

Peer-To-Peer Lending:

- P2P lending platforms connect borrowers with investors, offering better returns compared to traditional loan markets.

Deposits and payment services

Role of deposits, types, history of payment services, SEPA, PSD2, fintech innovations, deposit insurance

- Platforms like Funding Circle and Prosper Marketplace facilitate borrowing and lending between individuals and businesses.
- P2P lending transforms the lending environment by providing alternative financing options for borrowers.